Literature Review
Introduction

The exponential growth of structured data in relational databases has intensified the need for intuitive data access methods. Traditional SQL expertise remains a significant barrier for non-technical users, creating operational inefficiencies. Large Language Models (LLMs) offer transformative potential in bridging this gap by translating natural language queries into SQL. This review synthesizes research on LLM-based Text-to-SQL systems, focusing on architectural frameworks, model performance, evaluation benchmarks, and real-world implementation challenges.

Methodology

The search was conducted using the following keywords: **“Text-to-SQL transformer models”, “LLM for SQL generation”**, **“Spider dataset”.** Databases used included **“PubMed”** and **“Google Scholar”.** The research was not filtered by date or type.

Inclusion and Exclusion Criteria

In conducting the literature review, inclusion and exclusion criteria were applied to ensure that only studies directly relevant to Text-to-SQL were considered. The criteria were defined to capture works that provide methodological, evaluative, or application-driven contributions to the field, while excluding papers that focus broadly on language models or theoretical aspects without a clear connection to Text-to-SQL.

Inclusion Criteria:

Studies were included if they:

- Focused on **methods, systems, or frameworks for translating natural language to SQL**.

- Evaluated **LLM-based Text-to-SQL approaches** on benchmark or domain-specific datasets.
- Introduced **new datasets, benchmarks, or evaluation methods** for Text-to-SQL.
- Presented **survey papers** dedicated to Text-to-SQL.
- Applied Text-to-SQL in **specific domains or real-world scenarios** (e.g., healthcare, IoT, software engineering).

Exclusion Criteria:

Studies were excluded if they:

- Addressed **general LLM topics** (e.g., scaling laws, fine-tuning) without connection to Text-to-SQL.
- Focused on tasks **other than natural language to SQL generation**, such as SQL debugging, SQL equivalence checking, or general query optimization.
- Concentrated on **theoretical AI concepts** with no relevance to SQL generation.
- Focused on **multimodal, vision, or unrelated NLP tasks**.

No	Article	Authors	Published	Link	Keywords	Conclusions	Datasets	Approach/Architecture	Accuracy
1	Next-Generation Database Interfaces: A Survey of LLM-	Zijin Hong, Zheng Yuan, Qinggang Zhang, Hao Chen, Junnan Dong, Feiran Huang,	2025	Link	Text-to-sql, LLM	This survey reviews how large language models have transformed text-to-SQL tasks, highlighting key techniques, datasets, and challenges. It emphasizes recent LLM-based methods that improve query accuracy and discusses future directions for making these systems more robust and explainable.		Comparative Study	

	based Text-to-SQL	Xiao Huang							
2	From Natural Language to SQL: Review of LLM-based Text-to-SQL Systems	Ali Mohammadjafari, Anthony S. Maida, Raju Gottumukkala	2025	Link	Text-to-sql, LLM	The paper surveys advances in translating natural language queries into SQL using large language models, tracking progress from early rule-based systems to modern LLM-powered solutions, and underlining their impact on database accessibility.		Comparative Study	
3	SQLfuse: Enhancing Text-to-SQL Performance through Comprehensive LLM Synergy	Tingkai Zhang, Chaoyu Chen, Cong Liao, Jun Wang, Xudong Zhao, Hang Yu, Jianchao Wang, Jianguo Li, Wenhui Shi	2024	Link	Text-to-sql, LLM	Text-to-SQL simplifies database querying by enabling users to ask questions in natural language. This approach expands access to data analysis, making it easier for non-experts to interact with complex databases.	Spider	SQL mining, SQL linking, SQL generation, SQL critic, Fine-tuning with open-source LLMs	Test Acc : 85.6
4	Decomposition for Enhancing Attention: Improving LLM-based Text-to-SQL through Workflow Paradigm	Yuanzhen Xie, Xinzhou Jin, Tao Xie, MingXiong Lin, Liang Chen, Chenyun Yu, Lei Cheng, ChengXiang Zhuo, Bo Hu, Zang Li	2024	Link	Text-to-sql, LLM	To overcome limitations in single-step LLM prompting for text-to-SQL, the authors propose a workflow combining decomposition, self-correction, and active learning. This method enhances LLM performance and sets new benchmarks on standard datasets.	Spider, Spider Realistic, BIRD	Information Determination, Classification&&Hint, SQL Generation, Self-Correction, Active Learning	Dev Acc: 85.4 Test Acc: 87.1
5	Knowledge-to-SQL: Enhancing SQL Generation with Data Expert LLM	Zijin Hong, Zheng Yuan, Hao Chen, Qinggang Zhang, Feiran Huang, Xiao Huang	2024	Link	Text-to-sql, LLM	The paper introduces a framework that supplements missing domain knowledge in database schemas with a specialized language model, improving the accuracy and robustness of text-to-SQL generation beyond what standard LLMs achieve.	BIRD, Spider	The Supervised fine-tuned model, The preference learning framework, An off-the-shelf text-to-SQL model (Main purpose is to assist a Text-to-SQL model)	Approximately: %5 improvement on EX, %6 improvement on VES

6	LLM-Based Text-to-SQL for Real-World Databases	Eduardo R. Nascimento, Grettel García, Yenier T. Izqueirdo, Lucas Feijó, Gustavo M. C. Coelho, Aiko R. de Oliveira, Melissa Lemos, Robinson L. S. Garcia, Luiz A. P. Paes Leme & Marco A. Casanova	2024	Link	Text-to-sql, LLM	This study reveals that existing LLM-based text-to-SQL tools struggle with complex real-world databases due to schema representation issues. It proposes LLM-friendly views and data descriptions aligned with user language to significantly improve query generation performance.	IndDB, Mondial	Should take a look at Evaluation Procedure Section (Generates a real world database and tests Text-to-SQL Models)	
7	Text-to-SQL Empowered by Large Language Models: A Benchmark Evaluation	Dawei Gao, Haibin Wang, Yaliang Li, Xiuyu Sun, Yichen Qian, Bolin Ding, Jingren Zhou	2023	Link	Text-to-sql, LLM	The authors systematically compare various prompt engineering techniques for LLM-based text-to-SQL systems—examining question representation, example selection, and organization strategies. Based on these insights, they introduce **DAIL-SQL**, which achieves a new state-of-the-art execution accuracy of **86.6 %** on the Spider benchmark	Spider, Spider-Realistic	DAIL-SQL (LLM-based prompt engineering with Code Representation, DAIL Selection, and DAIL Organization)	Spider Test Set Accuracy: 86.6
8	Knapsack Optimization-based Schema Linking for LLM-based Text-to-SQL Generation	Zheng Yuan, Hao Chen, Zijin Hong, Qinggang Zhang, Feiran Huang, Qing Li, Xiao Huang	2025	Link	Text-to-sql, LLM	This paper addresses the critical schema linking step in text-to-SQL systems, which often suffers from missing relevant tables or including redundant ones. The authors introduce **KaSLA**, a plug-in method using knapsack optimization and a hierarchical linking strategy to accurately select tables and columns. Their enhanced schema linking metric and approach significantly boost SQL generation performance on benchmarks like Spider and BIRD, outperforming prior state-of-the-art models	Spider, BIRD	KaSLA (Knapsack Optimization-based Schema Linking)	Improves SOTA models by 2.47–3.20% on BIRD-dev; best result: 64.73% EX with RSL-SQL + KaSLA
9	TURSpider: A Turkish Text-to-SQL Dataset and LLM-Based	Ali Bugra Kanburoglu, Faik Boray Tek	2024	Link	Test-to-sql, LLM, Spider	This paper presents TURSpider, the first large-scale Turkish Text-to-SQL dataset, adapted from Spider. It benchmarks LLMs for Turkish SQL generation, introducing a feedback mechanism and a multi-agent approach (MoAF-SQL) to improve performance and robustness.	TURSpider (Turkish-translated Spider)	Fine-tuning Turkish LLMs (Turkcell, Trendyol, Sambalingo) + Chain-of-Feedback (CoF)	Best: 60.05% (TurkcellSQL + CoF) on TURSpider-dev

	Study								
10	Large Language Model Enhanced Text-to-SQL Generation: A Survey	Xiaohu Zhu, Qian Li, Lizhen Cui, Yongkang Liu	2024	Link	Sql, LLM, Survey	The authors provide a comprehensive survey of text-to-SQL generation approaches enhanced by LLMs, categorizing methods into prompt engineering, fine-tuning, task-specific training, and agent-based systems. They also offer a detailed review of datasets and evaluation metrics. This work aims to clarify current research patterns, identify challenges, and guide future developments in LLM-powered text-to-SQL systems		Comparative Study	
11	A Survey on Employing Large Language Models for Text-to-SQL Tasks	Liang Shi, Zhengju Tang, Nan Zhang, Xiaotong Zhang, Zhi Yang	2025	Link	Sql, LLM, Survey	This survey reviews over 170 studies on using LLMs for text-to-SQL, comparing prompting and fine-tuning methods. It outlines key challenges like schema linking and domain adaptation, offering guidance for future research		Comparative Study	
12	A Survey of Large Language Model-Based Generative AI for Text-to-SQL: Benchmarks, Applications, Use Cases, and Challenges	Aditi Singh, Akash Shetty, Abul Ehtesham, Saket Kumar, Tala Talaei Khoei	2025	Link	Sql, LLM, Survey	This survey reviews recent LLM-based approaches to text-to-SQL, categorizing works by prompt engineering, fine-tuning, task-specific training, and agent-based methods. It synthesizes insights from over 170 studies, highlights prevailing challenges such as schema linking and domain adaptation, and outlines future research directions in efficiency, accuracy, and robustness		Comparative Study	
13	A Survey of Text-to-SQL in the Era of LLMs: Where are We, and Where are We	Xinyu Liu, Shuyu Shen; Boyan Li, Peixian Ma, Runzhi Jiang, Yuxin Zhang	2025	Link	Sql, LLM, Survey	This survey reviews LLM-based text-to-SQL systems, presenting a modular taxonomy, error classification, and practical guidance. It highlights key challenges like generalization, efficiency, and robustness for future work.		Comparative Study	

	Going?								
14	Exploring the Landscape of Text-to-SQL with Large Language Models: Progresses, Challenges and Opportunities	Yiming Huang, Jiyu Guo, Wenxin Mao, Cuiyun Gao, Peiyi Han, Chuanyi Liu, Qing Ling	2025	Link	Sql, LLM	This systematic survey examines LLM-based text-to-SQL research, focusing on trends, techniques, datasets, evaluation metrics, and challenges. It offers a clear roadmap for future work by highlighting gaps and opportunities in schema linking, error analysis, and domain adaptation		Comparative Study	
15	Enhancing LLM Fine-tuning for Text-to-SQLs by SQL Quality Measurement	Shouvon Sarker, Xishuang Dong, Xiangfang Li, Lijun Qian	2024	Link	Sql, LLM	The authors propose a novel feedback mechanism that evaluates generated SQL based on quality and execution results, then uses this to fine-tune LLMs. Tested on the BIRD benchmark, it achieves execution accuracy and efficiency comparable to state-of-the-art models like GPT-4 and T5	BIRD	A method combining Human-Designed Step-by-Step Prompts (HDSP) and a SQL Quality Measurement feedback loop. The feedback uses normalized (syntactic) and semantic comparisons to classify SQL difficulty and dynamically enhance the prompts.	EX: %49.87 VES: %47.32
16	Optimizing text-to-SQL conversion techniques through the integration of intelligent agents and large language models	Samuel Ojuri , The Anh Han , Raymond Chiong , Alessandro Di Stefano	2023	Link	Sql, LLM	This study combines intelligent agents with LLMs to help non-technical users generate accurate SQL queries. Fine-tuned examples significantly improve performance, enabling faster and easier data access.	MySQL Classicmodels sample database (domain-specific; curated NLQ–SQL pairs, plus synthetic data from GPT-4)	Integration of Intelligent Agents with LLMs (ReAct framework) - Compared Fine-tuning vs. Few-shot In-context Learning - Models tested: GPT-3.5-turbo (fine-tuned & few-shot), Meta-Llama-3-8B-Instruct (fine-tuned & few-shot), GPT-4, Llama-3.3-70B-Instruct-Turbo	est models: • GPT-4: Valid SQL 99%, Execution Accuracy 95.5%, Test-Suite Accuracy 82% • Llama-3.3-70B: Valid SQL 99%, EX 95%, TS 81.5%
17	Conversion of natural language text to SQL	Arnav Jha, Naman Anand, H. Karthikeyan	2025	Link	Sql, LLM	This work introduces **Semantic Synthesis**, a method that converts user-provided natural language into precise SQL queries using a deep integration of LLMs and transformers. It targets non-technical users by providing	Not Specified	A Generative AI model with a Sequence-to-Sequence (Seq2Seq) architecture, enhanced with an attention	Not Reported

	queries using generative AI					context-aware SQL generation across domains like healthcare, e-commerce, education, and HR, paired with algorithmic transparency for the translation process. It aims to democratize database access via an intuitive interface that balances accuracy and interpretability		mechanism	
18	Text-to-SQL Domain Adaptation via Human-LLM Collaborative Data Annotation	Yuan Tian, Daniel Lee, Fei Wu, Tung Mai, Kun Qian, Siddhartha Sahai, Tianyi Zhang, Yunyao Li	2025	Link	Text-to-sql, LLM,survey	The paper introduces **SQLsynth**, a human-in-the-loop annotation system that combines experts and LLMs to efficiently generate domain-specific text-to-SQL training data. In user studies, SQLsynth reduces cognitive load, speeds up annotation, and produces more accurate and diverse datasets compared to manual annotation or using ChatGPT alone	Spider	SQLsynth: A human-in-the-loop system combining PCFG-based SQL generation + LLM-based NL translation + interactive error detection & repair.	%95.56
19	LLM driven Natural Language to SQL for Utility Dashboards	Jyotirmoy Banerji; Abhiraj Amin; Kapil Rathor	2025	Link	Sql, LLM	This paper introduces an NL2SQL system optimized for IoT plant data, using fine-tuned LLMs and custom context structures. It improves SQL generation accuracy for complex, context-rich IoT queries and is evaluated with the Spider dataset.	Spider, Custom IoT SQL dataset	Fine-tuned GPT-2 with a custom tag datasheet for IoT context + ChatGPT API for evaluation	Precision: 0.89, Recall: 0.87, F1 Score: 0.88
20	Analyzing the Effectiveness of Large Language Models on Text-to-SQL Synthesis	Richard Roberson, Gowtham Kaki, Ashutosh Trivedi	2024	Link	Sql, LLM, Survey	This study compares fine-tuning an open-source model (WizardCoder-15B) with a pipeline using GPT-3.5-Turbo plus GPT-4 for error correction. WizardCoder achieves ~61% execution accuracy, while the GPT-based pipeline reaches ~82%. The authors analyze common mistakes in generated SQL—such as wrong columns, join errors, and misordered aggregations—offering insights for future improvements.	Spider	Two-pronged: 1) Fine-tuned open-source WizardCoder-15B; 2) Fine-tuned GPT-3.5-Turbo-16k + GPT-4-Turbo for error correction	Overall Execution Accuracy: 82.1% (Easy: 94.0%, Medium: 85.7%, Hard: 80.5%, Extra Hard: 56.6%)
~~21~~	~~SWE-SQL: Illuminating LLM Pathways to Solve User SQL Issues in Real-World Applications~~	Jinyang Li, Xiaolong Li, Ge Qu, Per Jacobsson, Bowen Qin, Binyuan Hui, Shuzheng Si, Nan Huo, Xiaohan Xu, Yue Zhang, Ziwei Tang, Yuanshuai	2025	Link	Sql, LLM	This work introduces **BIRD-CRITIC**, a new benchmark comprising 530 PostgreSQL-only and 570 multi-dialect SQL debugging tasks sourced from real-world user issues. Standard LLMs achieve only ~39% success. To tackle this, the authors propose **Six-Gym**—an environment that reverse-engineers SQL problems (SQL-Rewind) and leverages **f-Plan Boosting** to train models more effectively. Their agent **Bird-Fixer**, built on Qwen-2.5-Coder-14B, outperforms leading proprietary models on realistic SQL-debugging tasks			

		Li, Florensia Widjaja, Xintong Zhu, Feige Zhou, Yongfeng Huang, Yannis Papakonstantinou, Fatma Ozcan, Chenhao Ma, Reynold Cheng						
~~22~~	~~LLM-SQL-Solver: Can LLMs Determine SQL Equivalence?~~	Fuheng Zhao, Jiayue Chen, Lawrence Lim, Ishtiyaque Ahmad, Divyakant Agrawal, Amr El Abbadi	2023	Link	Sql, LLM	This paper explores using LLMs to judge if two SQL queries are semantically equivalent. It proposes prompt-based methods that show strong alignment with human judgment, offering a promising alternative to traditional evaluation metrics.			
23	Improving Text-to-SQL with a Hybrid Decoding Method	Geunyeong Jeong , Mirae Han , Seulgi Kim , Yejin Lee , Joosang Lee , Seongsik Park , Harksoo Kim	2023	Link	Text-to-sql	This study proposes a hybrid decoder that combines sketch-based and generation-based methods for Text-to-SQL, improving syntactic accuracy and efficiency. A value-prediction module for `WHERE` clauses and a new syntax-error evaluation metric further enhance performance.	WikiSQL	Hybrid Decoder (combines sketch-based and generation-based methods) + Value Prediction Module using sequence labeling	Logical Form Accuracy: 83.5%, Execution Accuracy: 89.1%, Syntactic Error Rate: 0.00%
~~24~~	~~Redefining text-to-SQL metrics by incorporating semantic and structural similarity~~	Giovanni Pinna , Yuriy Perezhohin , Luca Manzoni , Mauro Castelli , Andrea De Lorenzo	2025	Link	Text-to-sql	This study introduces a new evaluation metric that more accurately measures SQL query similarity—going beyond simple binary or exact-match scoring by factoring in structural differences, partial correctness, and semantic alignment. The metric allows finer-grained evaluation and better ranking of text-to-SQL models, aiding both model assessment and development.			
25	Towards Understanding the	Richard Tarbell, Kim-Kwang Raymond Choo, Glenn Dietrich,	2024	Link	Text-to-sql	The study shows that while current models perform well on existing medical Text-to-SQL benchmarks (with accuracies over 90%), their performance drops sharply (to around 28%) when evaluated on more generalizable	MIMICSQL, MIMICSQL 2.0, Spider	Base Model: T5 (a Text-to-Text Transformer), fine-tuned for text-to-SQL. Data Augmentation: Back-	Original Split (P→P): 93.6%, New Split (P→P):

	Generalization of Medical Text-to-SQL Models and Datasets	Anthony Rios				splits. The authors introduce a data augmentation technique to improve robustness and highlight that the medical domain still needs more reliable Text-to-SQL solutions		translation (to/from French & German) to generate synthetic paraphrases. Training Strategy: Explored single-domain (MIMICSQL) and multi-domain (MIMICSQL + Spider) training.	~52.8%, New Split (T→P): ~50.1%
26	A Heterogeneous Graph to Abstract Syntax Tree Framework for Text-to-SQL	Ruisheng Cao, Lu Chen, Jieyu Li, Hanchong Zhang, Hongshen Xu, Wangyou Zhang, Kai Yu	2023	Link	Text-to-sql	The authors propose **HG2AST**, which encodes input as a heterogeneous graph and decodes into SQL via an abstract syntax tree (AST). On the encoder side, **LGESQL** performs dual graph message passing; on the decoder side, a grammar-based structure with a learning algorithm controls node expansion order to reduce bias. Together, this unified framework achieves superior performance across diverse text-to-SQL datasets.	Spider, DuSQL, CSpider, WikiSQL,	Framework Name: HG2AST (Heterogeneous Graph to Abstract Syntax Tree). Encoder: LGESQL (Line Graph Enhanced Encoder) that performs dual message passing on the original node graph and a derived line graph to capture higher-order semantics. Decoder: A grammar-based decoder that constructs a SQL Abstract Syntax Tree (AST). Key Innovation: Golden Tree-oriented Learning (GTL) algorithm to make the decoder order-insensitive and avoid permutation bias during AST generation.	Spider (EM): 75.1%, DuSQL (EMV): 77.4%, CSpider (EM): 57.0%
27	ExSPIN: Explicit Feedback-Based Self-Play Fine-Tuning for Text-to-SQL Parsing	Liang Yan, Jinhang Su, Chuanyi Liu, Shaoming Duan, Yuhao Zhang, Jianhang Li, Peiyi Han, Ye Liu	2025	Link	Text-to-sql	Introduces **ExSPIN**, a self-play fine-tuning method that incorporates explicit SQL execution feedback and schema hints, significantly improving performance on real-world text-to-SQL datasets.	Spider, BIRD	Framework Name: ExSPIN (Explicit Feedback-based Self-Play Fine-Tuning). Key Innovations: 1. Explicit Schema Integration, 2. Execution Feedback Fine-Tuning	Spider: %83.2 BIRD: %52.1
28	An interaction-modeling mechanism for context-dependent Text-	Wei Yu, Tao Chang, Xiaoting Guo, Mengzhu Wang, Xiaodong Wang	2021	Link	Text-to-sql	The authors propose a multi-turn, context-aware Text-to-SQL model where each input (natural language question, previous SQL, and database schema) is represented as a separate graph. They then aggregate these via heterogeneous graph aggregation into a unified representation, resulting in competitive performance on	SparC, CoSQL	HGASQL: A model using an interaction-modeling mechanism based on heterogeneous graph aggregation. It represents questions, SQL queries, and	SparC: Question Match: 51.5%, Interaction Match: 31.6% CoSQL:

	to-SQL translation based on heterogeneous graph aggregation					SparC and CoSQL benchmarks.		database schemas as individual graphs, then aggregates them into a holistic representation for SQL generation.	Question Match: 44.9%, Interaction Match: 16.2%
29	SEOSS-Queries - a software engineering dataset for text-to-SQL and question answering tasks	Mihaela Todorova Tomova, Martin Hofmann, Patrick Mäder	2022	Link	Text-to-sql	Introduces **SEOSS-Queries**, a software engineering–focused Text-to-SQL dataset containing 1,162 natural language utterances mapped to 166 SQL queries. Each query includes four precise and three more general paraphrased utterances, plus over 393,000 labeled comments extracted from issue trackers. The dataset provides SQLNet and RatSQL baselines for benchmarking.	SEOSS	Baseline models: SQLNet and RatSQL (with both GloVe and BERT embeddings). The study evaluates the dataset's validity and applicability using these established text-to-SQL models.	Best Exact Match Accuracy (RatSQL + GloVe): • 78.9% (Experiment 2, All queries) • 82.7% (Experiment 4, Non-specific utterances subset)
30	MedT5SQL: a transformers-based large language model for text-to-SQL conversion in the healthcare domain	Alaa Marshan, Anwar Nais Almutairi, Athina Ioannou, David Bell, Asmat Monaghan, Mahir Arzoky	2024	Link	Text-to-sql	Introduces **MedT5SQL**, a fine-tuned T5 model trained on the MIMICSQL healthcare dataset. It demonstrates effective natural language to SQL conversion in the healthcare domain, showing improved performance with additional training epochs.	MIMICSQL, WikiSQL	MedT5SQL: A fine-tuned T5-base transformer model (Text-to-Text Transfer Transformer)	On MIMICSQL: • 80.63% Exact Match Accuracy • 98.937% Approximate String Matching • 90% Manual Evaluation On WikiSQL: • 44.2% Exact Match Accuracy
31	DIN-SQL: Decomposed In-Context Learning of Text-to-SQL with Self-Correction	M. Pourreza, Davood Rafiei	2023	Link	Text-to-SQL	It is shown that SQL queries, despite their declarative structure, can be broken down into sub-Problems and the solutions of those sub-problems can be fed into LLMs to significantly improve their performance.	Spider, BIRD	Schema Linking, Classification & Decomposition, SQL Generation(Easy, Nested Complex, Non-Nested Complex), Self-correction	EX: %85.3 EM: %60

32	Spider: A Large-Scale Human-Labeled Dataset for Complex and Cross-Domain Semantic Parsing and Text-to-SQL Task	Tao Yu, Rui Zhang, Kai-Chou Yang, Michihiro Yasunaga, Dongxu Wang, Zifan Li, James Ma, Irene Z Li, Qingning Yao, Shanelle Roman, Zilin Zhang, Dragomir R. Radev	2018	Link	Text-to-SQL	This work defines a new complex and cross-domain semantic parsing and text-to-SQL task so that different complicated SQL queries and databases appear in train and test sets and experiments with various state-of-the-art models show that Spider presents a strong challenge for future research	Spider	Spider Dataset by Yale University	
33	CodeS: Towards Building Open-source Language Models for Text-to-SQL	Haoyang Li, Jing Zhang, Hanbing Liu, Ju Fan, Xiaokang Zhang, Jun Zhu, Renjie Wei, Hongyan Pan, Cuiping Li, Hong Chen	2024	Link	Text-to-SQL	This paper studies the research challenges in building CodeS, a fully open-source language model, which achieves superior accuracy with much smaller parameter sizes, and addresses the challenges of schema linking and rapid domain adaptation through strategic prompt construction and a bi-directional data augmentation technique	Spider(DK, SYN, Realistic,Dr. Spider), BIRD	Incremental pre-training, Database prompt construction, Bi-directional augmentation for new domain adaptation, Supervised fine-tuning of CodeS, Few-shot in-context learning of CodeS	Spider Accuracy (EX): - CodeS-1B: 77.9% - CodeS-3B: 83.4% **- CodeS-7B: 85.4%** - CodeS-15B: 84.9%
34	RAT-SQL: Relation-Aware Schema Encoding and Linking for Text-to-SQL Parsers	Bailin Wang, Richard Shin, Xiaodong Liu, Oleksandr Polozov, Matthew Richardson	2019	Link	Text-to-SQL	This work presents a unified framework, based on the relation-aware self-attention mechanism, to address schema encoding, schema linking, and feature representation within a text-to-SQL encoder and achieves the new state-of-the-art performance on the Spider leaderboard.	Spider, WikiSQL	Encoder-Decoder Structure: Encoder:GloVe + BiLSTM + Relation-Aware Sefl-Attention(RAT), Decoder: AST-based LSTM decoder Relation types: table/column relationships + question-schema links RAT-SQL + BERT variant was also tested	Accuracy (Spider) - RAT-SQL: 57.2% (test) - RAT-SQL + BERT: 65.6% (test) - RAT-SQL + BERT: 69.7% (dev) Accuracy (WikiSQL) - Execution Accuracy: ~79.5% (dev) - Logical Form

									Accuracy: ~73.3% (test)
35	RESDSQL: Decoupling Schema Linking and Skeleton Parsing for Text-to-SQL	Haoyang Li, Jing Zhang, Cuiping Li, Hong Chen	2023	Link	Text-to-SQL	This paper proposes a ranking-enhanced encoding and skeleton-aware decoding framework to decouple the schema linking and the skeleton parsing at Text-to-SQL and evaluates the proposed framework on Spider and its three robustness variants: Spider-DK, Spider-Syn, and Spider-Realistic.	Spider, Spider-DK, Spider-Syn, Spider-Realistic	- Based on T5 (Seq2Seq Transformer) - Cross-encoder ranks relevant schema - Skeleton-aware decoder: first SQL structure, then details - Uses NatSQL (intermediate representation)	Accuracy: Spider (Dev): -RESDSQL-3B: EM 78.0%, EX 81.8% - RESDSQL-3B + NatSQL: EM 80.5%, EX 84.1%
36	Spider 2.0: Evaluating Language Models on Real-World Enterprise Text-to-SQL Workflows	Fangyu Lei, Fangyu Lei, Jixuan Chen, Yuxiao Ye, Ruisheng Cao, Dongchan Shin, Hongjin Su, Zhaoqing Suo, Hongcheng Gao, Hongcheng Gao, Pengcheng Yin, Victor Zhong, Caiming Xiong, Ruoxi Sun, Qian Liu, Sida Wang, Tao Yu	2024	Link	Text-to-SQL	This work introduces Spider 2.0, an evaluation framework comprising 632 real-world text-to-SQL workflow problems derived from enterprise-level database use cases, and shows that while language models have demonstrated remarkable performance in code generation, they require significant improvement in order to achieve adequate performance for real-world enterprise usage.	Spider 2.0	Spider 2.0 Dataset	
37	CHASE-SQL: Multi-Path Reasoning and Preference Optimized Candidate Selection in Text-to-SQL	Mohammadreza Pourreza, Hailong Li, Ruoxi Sun, Yeounoh Chung, Shayan Talaei, Gaurav Tarlok Kakkar, Yu Gan, Amin Saberi, Fatma Ozcan, Sercan Ö. Arik	2024	Link	Text-to-SQL	CHASE-SQL is introduced, a new framework that employs innovative strategies, using test-time compute in multi-agent modeling to improve candidate generation and selection, and achieves the state-of-the-art execution accuracy on the BIRD Text-to-SQL dataset benchmark	BIRD	- Multi-agent SQL generation (CoT, decomposition, synthetic examples) - LLM-based candidate selection via pairwise ranking	Accuracy - BIRD (Dev/Test): EX ~73.0%

38	MCS-SQL: Leveraging Multiple Prompts and Multiple-Choice Selection For Text-to-SQL Generation	Dongjun Lee, Choongwon Park, Jaehyuk Kim, Heesoo Park	2024	Link	Text-to-SQL	This study considers the sensitivity of LLMs to the prompts and introduces a novel approach that leverages multiple prompts to explore a broader search space for possible answers and effectively aggregate them.	Spider, BIRD	- In-context learning (ICL) method (using GPT-4) - Use multiple prompts for schema linking and SQL generation - Generate multiple candidate SQLs, then select best via multiple-choice selection (MCS) with confidence filtering	- BIRD: EX = 65.5%, VES = 71.4% - Spider (test): EX = 89.6%

21 → Not about text-to-sql
22 → Not about text-to-sql
24 → About measuring text-to-sql, not developing it.